

Subject Description Form

Subject Code	ABCT5031
Subject Title	Ecological Approaches for Carbon Management
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The subject aims to guide students through the application of ecological science in achieving carbon neutrality, thereby promoting the sustainable development of our modern society. We will explore the chemistry behind the global carbon cycle and climate change, and the role of ecosystems in regulating these environmental processes. This course aims to equip students with a comprehensive understanding of carbon management principles and ecological approaches, providing them with knowledge of various ecosystems involved in carbon sequestration, and updating them on the operating principles and market opportunities related to carbon management.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand, in a broad context, the basic knowledge of ecological approaches involved in carbon management, including carbon management principles and the global carbon cycle. (b) Measure and analyze greenhouse gases and calculate carbon footprints. (c) Assess the carbon sequestration potential of various ecosystems and evaluate urban carbon management practices. (d) Integrate ecological and technological approaches for comprehensive carbon management strategies.
Subject Synopsis/ Indicative Syllabus	• Fundamentals of Ecological Approaches and Carbon Management • Carbon Footprint Analysis - Understanding Carbon Footprint. - Reducing Strategies of Carbon Footprint. - Life cycle assessment (LCA). • Greenhouse gas emissions (GHG) capture, utilization, and storage - Ecological, biological, and chemical carbon sequestration - Overview of GHG capture, utilization, and storage - CO₂ capture, utilization, and storage (CCUS) - Corporate carbon audit quantifying the emissions

	• Carbon Sequestration in Forest and Soil Ecosystems. - Forests as Carbon Sinks. - Forest Management Practices. - Soil Carbon Dynamics. - Agricultural Practices for Soil Carbon Sequestration. - Monitoring and Measuring Soil Carbon. • Wetlands, Peatlands, and Marine Ecosystems in Carbon Management. - Wetlands and Peatlands as Carbon Reservoirs. - Marine and Coastal Carbon Sequestration. - Management and Conservation Strategies. • Urban Carbon Management, Integrative Approaches and Future Directions. - Urban Ecosystems and Carbon Sequestration. - Innovative Technologies in Urban Carbon Management - Sustainable Urban Planning. - Case studies.																																								
Teaching/Learning Methodology	Lectures: Key concepts and background information will be introduced during lectures. Students are expected to read and understand the textbook contents after the lectures, as it provides additional details. Supplementary reading materials from the literature will be provided before lectures. Guest speakers may be invited to lecture on specific topics as needed. The flipped classroom approach will be employed to foster interactive discussions and promote critical thinking. Midterm Exam: A midterm examination will be conducted to consolidate the students’ learning, identify areas that require further review, and prepare for the more advanced topics to follow. It will encourage consistent study habits and provide valuable feedback on their progress. Projects: Practical projects will be assigned to help students gain a deep understanding of the course material. These projects will provide hands-on experience and application of theoretical concepts. Group presentations: Students will work in groups to deliver presentations on selected topics. They will answer questions from the lecturer and their peers, reinforcing teamwork and collaborative learning. This will enhance their understanding of the topics and improve their communication and presentation skills.																																								
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">Percentage weighting</th><th colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. In-class activity</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Midterm Exam</td><td>30%</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>3. Presentation</td><td>25%</td><td></td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>4. Group Project</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td></td><td></td><td></td><td></td></tr></table>	Specific assessment methods/tasks	Percentage weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. In-class activity	20%	✓	✓	✓		2. Midterm Exam	30%	✓	✓			3. Presentation	25%		✓	✓	✓	4. Group Project	25%	✓	✓	✓	✓	Total	100 %				
Specific assessment methods/tasks	Percentage weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																																					
		a	b	c	d																																				
1. In-class activity	20%	✓	✓	✓																																					
2. Midterm Exam	30%	✓	✓																																						
3. Presentation	25%		✓	✓	✓																																				
4. Group Project	25%	✓	✓	✓	✓																																				
Total	100 %																																								

	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Each student will be required to participate in class activities, including a flipped classroom and group discussion, which will assess their ability to understand the related topic and critically think about the related issues. Students will be grouped to deliver presentations on selected topics, which will assess their ability to conduct case analysis, present the result, use proper skills for presentation, and argue points in support of their rationale. For the summative part, a mid-term exam will be used to evaluate students' level of understanding and provide higher-order thinking questions to assess students' analytical and problem-solving skills. The project will demonstrate the extent to which students have learned from the subject and provide practical opportunities related to carbon management.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	26 Hrs.
	▪ Tutorial	13 Hrs.
	Other student study efforts:	
	▪ Exercise	9 Hrs.
	▪ Self-study and preparation for the exam	36 Hrs.
	▪ Preparation for the group project and presentation	36 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	1. Carbon footprint analysis: concepts, methods, implementation, and case studies; Matthew John Franchetti, Defne Apul; CRC Press, Taylor& Francis Group; 2013 2. Carbon Management for a Sustainable Environment 1st ed. 2020 Edition; Shelley W. W. Zhou; Springer; 2020	